

SNS Performance and the Next Generation of High Power Proton Accelerators

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Outline

- **SNS experience, last 2 years**
 - The beginning: RFQ woes
 - The middle: Sensitive Superconducting RF Linac
 - The end: Broken targets
- **SNS Upgrade Plans**
- **Future high power accelerators**
 - Present, Upgrades and Future Facilities
 - (Snowmass 2013 Proton Capability working group)

SNS Accelerator Complex

Front-End:
Produce a 1-msec
long, chopped,
H⁻ beam

**1 GeV
LINAC**

**Accumulator
Ring:** Compress 1
msec long pulse to
700 nsec
1000 MeV

2.5 MeV

Front-End

LINAC

**Liquid
Hg
Target**

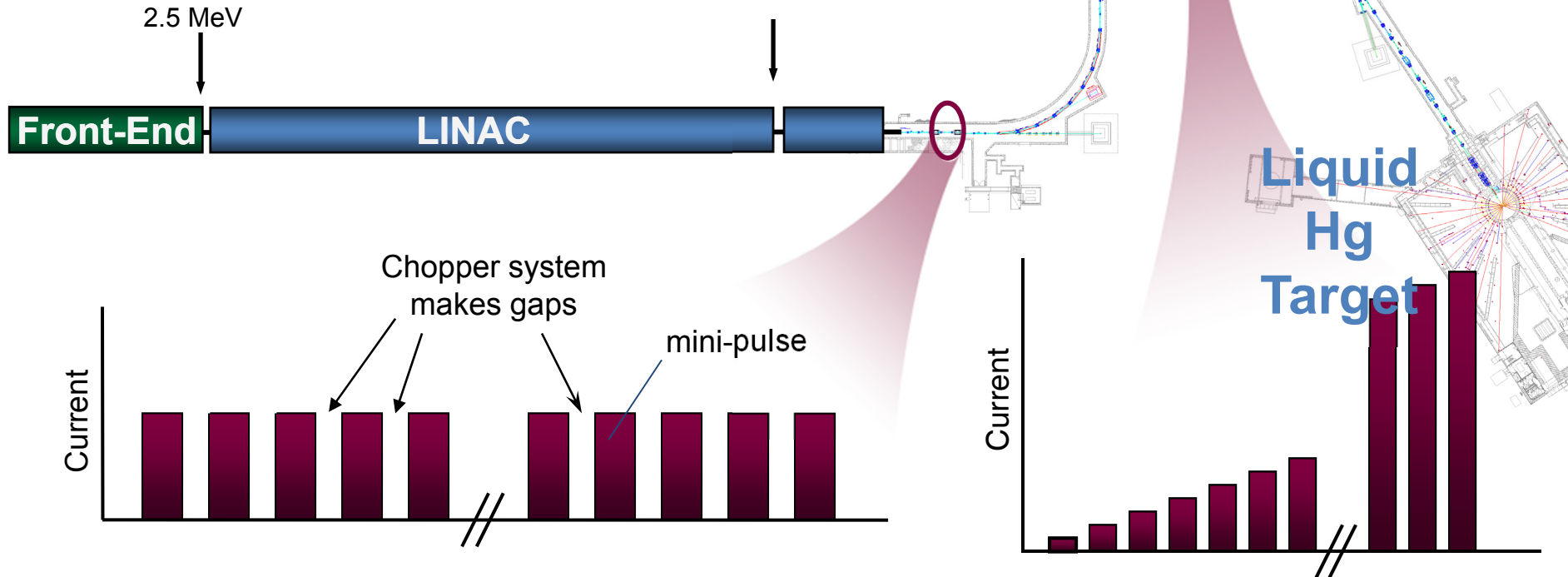
Neutron scattering users want beam power and reliability

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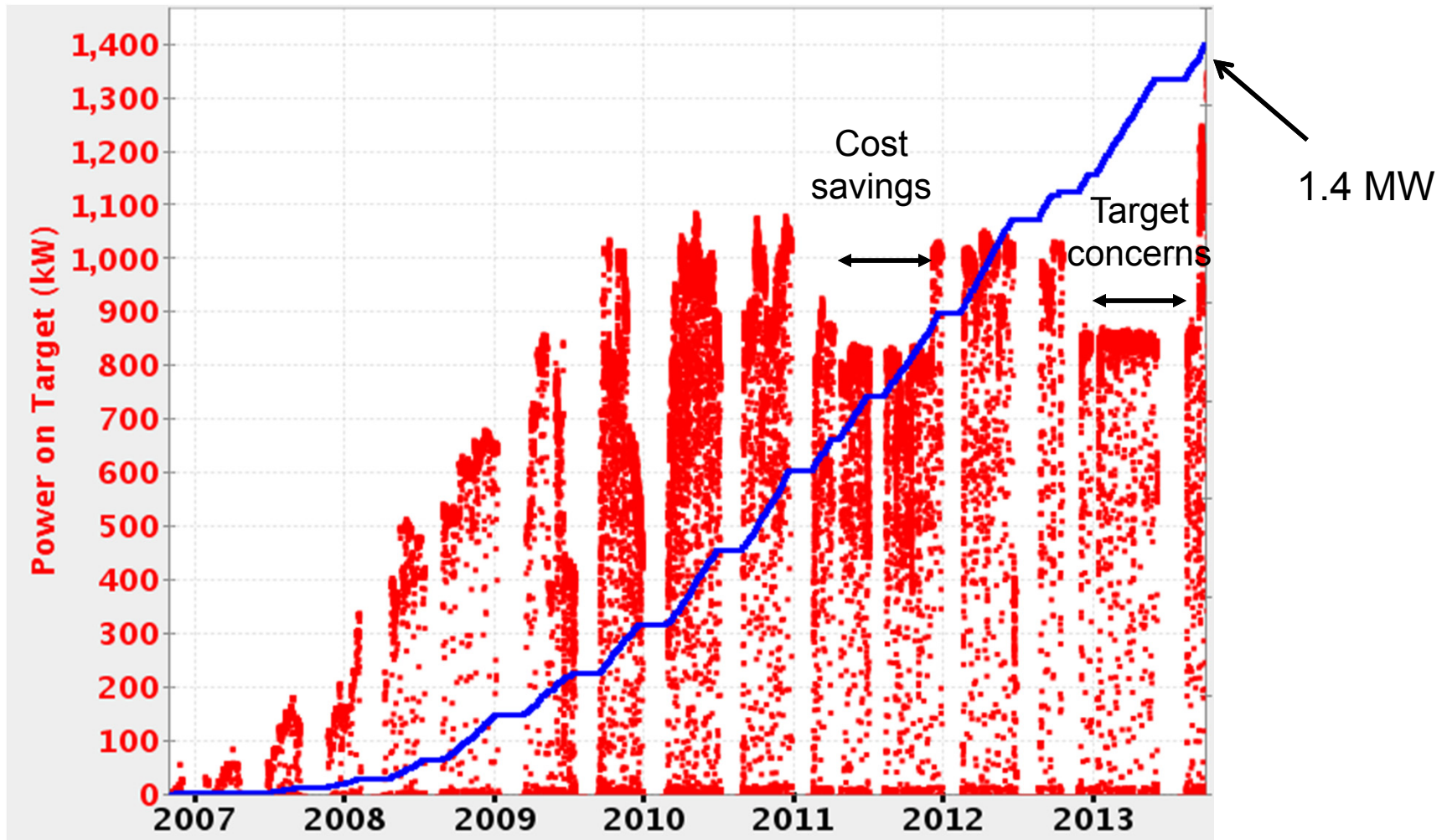
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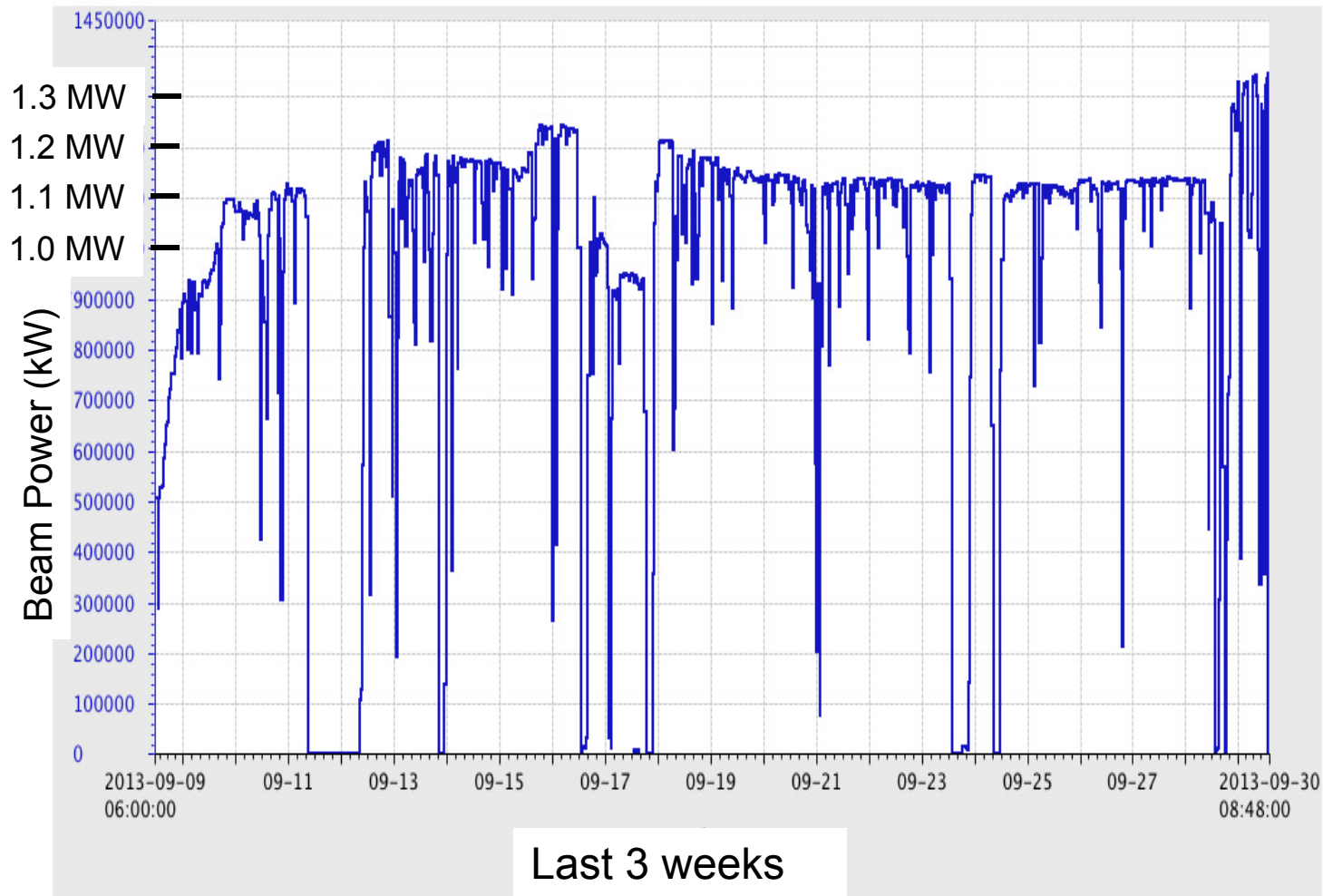
SNS Power History



- Last three years have some reduced power operation (850 kW)
 - Not accelerator limited
 - Recently operated up to 1.4 MW

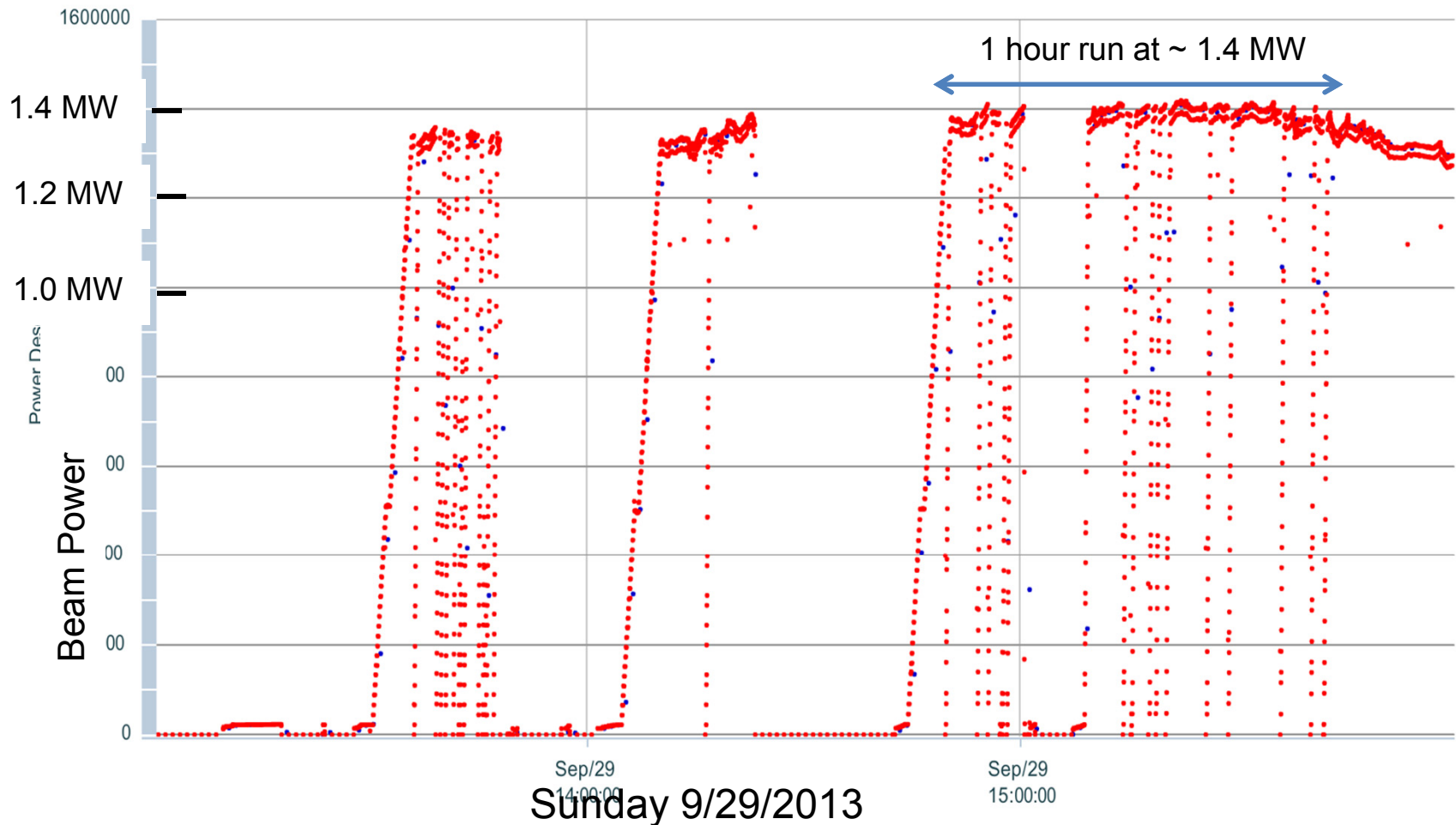
Sept. 2013: Operation at 1.1-1.2 MW

Brief Stint at 1.4 MW

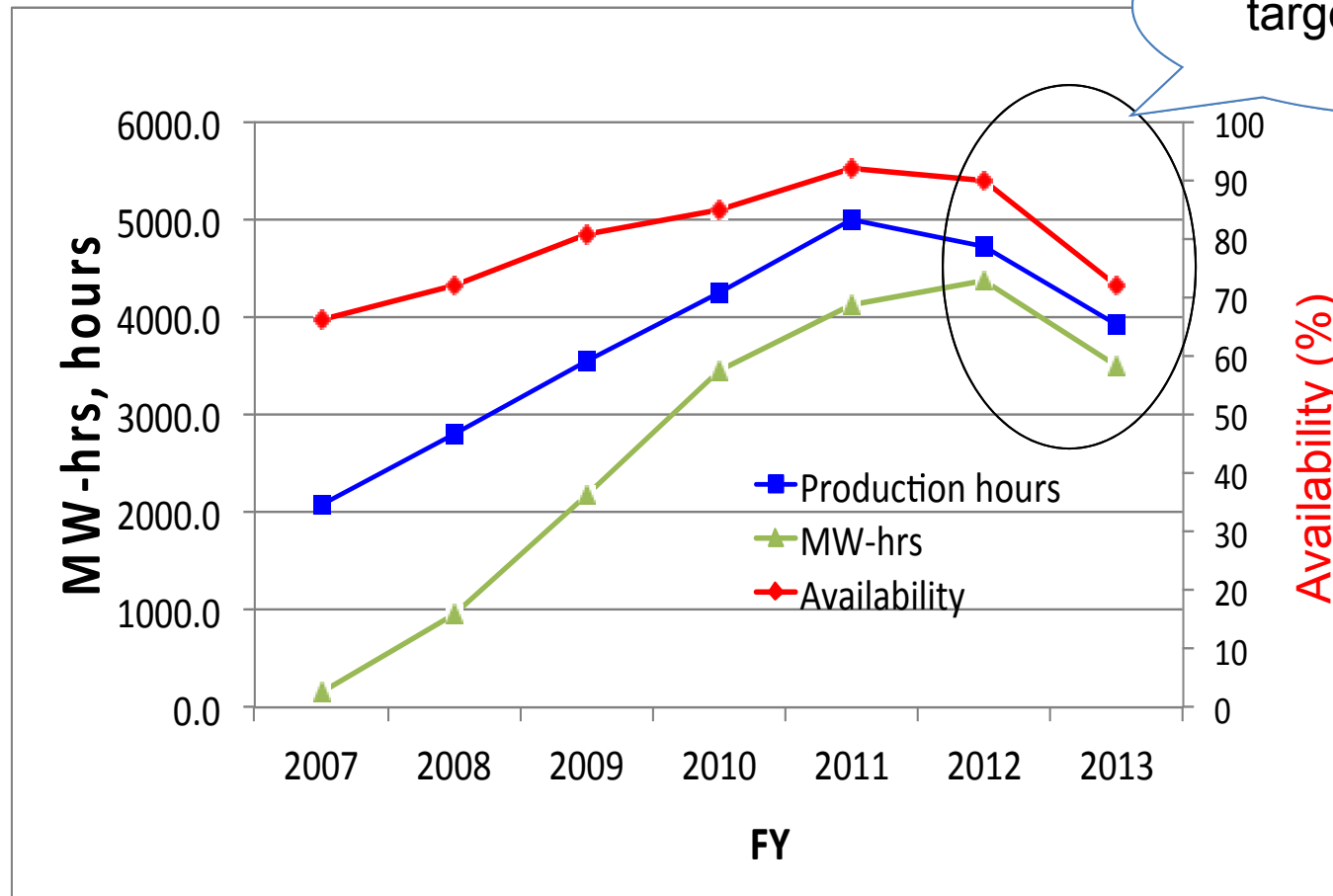


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High Level Facility Metrics



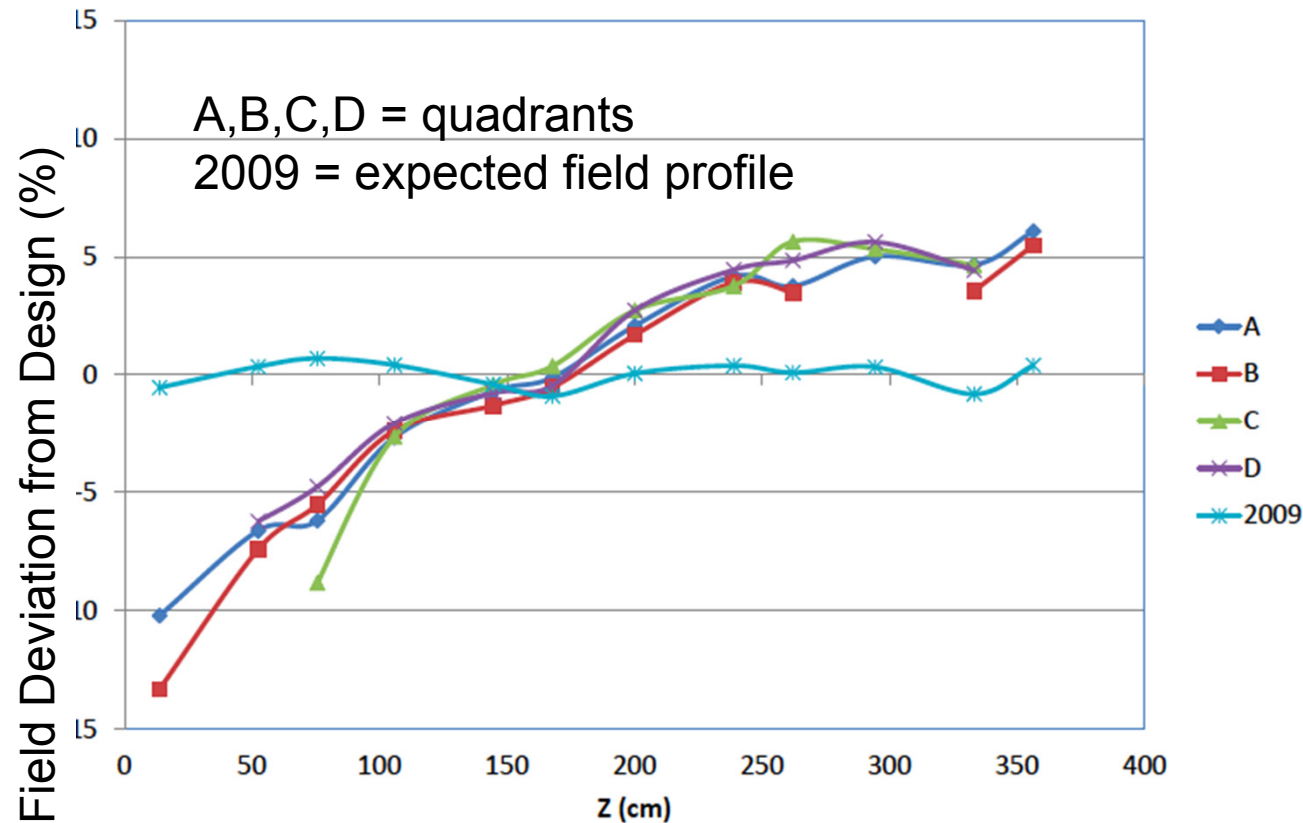
- Operational hours and reliability are important user facility metrics

— 2012 and 2013 figures are impacted by target failures

SNS RFQ: Unexplained Detuning Incidents

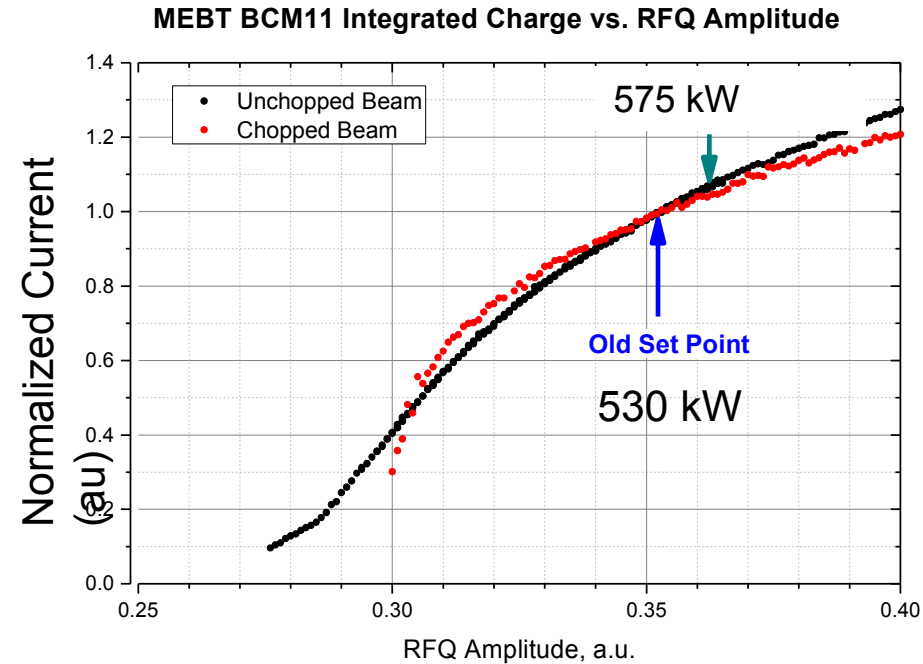
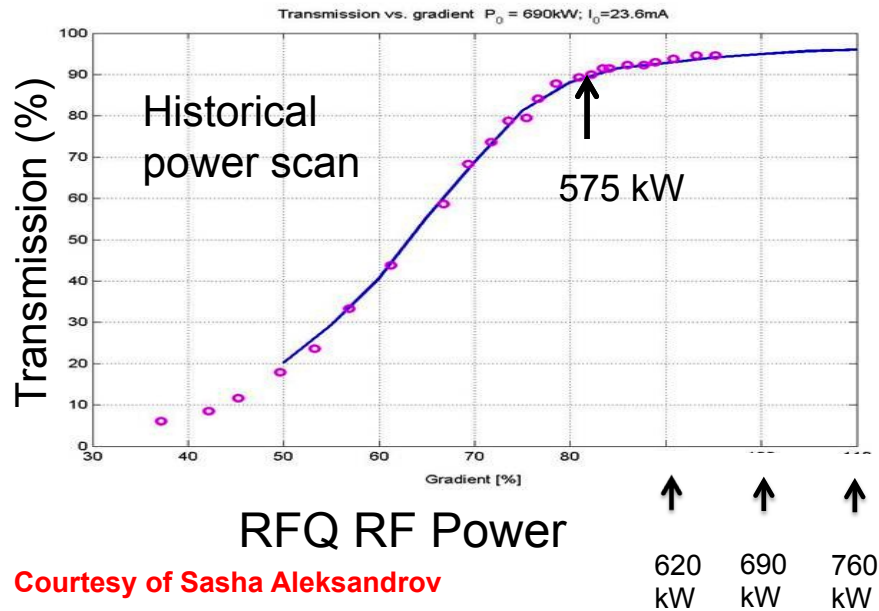
- **“Systematic reduction” of ion source performance noted mid 2011-2012**
- **Beam transmission measurements through RFQ indicated high beam loss in RFQ**
- **Examination of the RFQ field profile showed significant deviation from design**
 - **This is the third time a “detuning” incident has occurred**

RFQ Field Measurement early in late 2012 Indicate a Problem



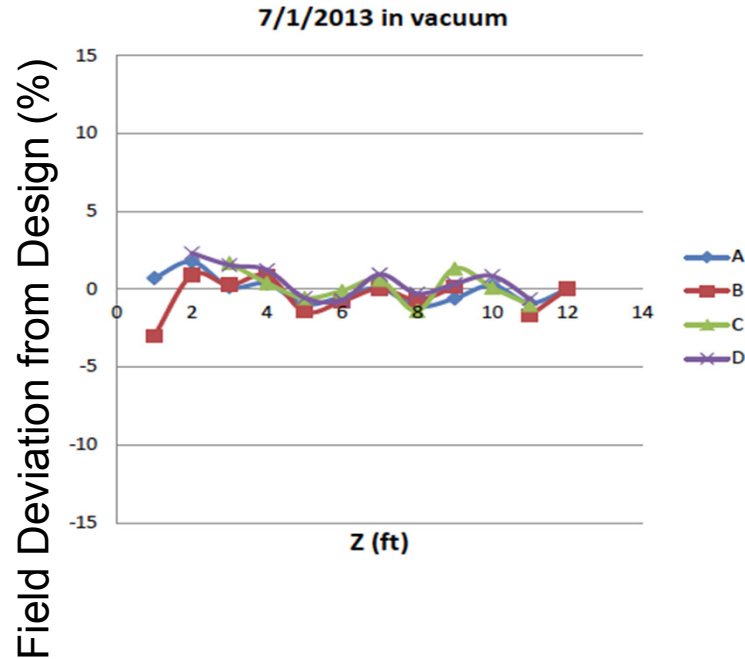
- Clear indication of field tilt from expectation

RFQ Power Scans Indicate a Problem



- We are operating significantly below “transmission saturation” RF power level
 - Resonance control (cooling) cannot support too large an input power

RFQ Status



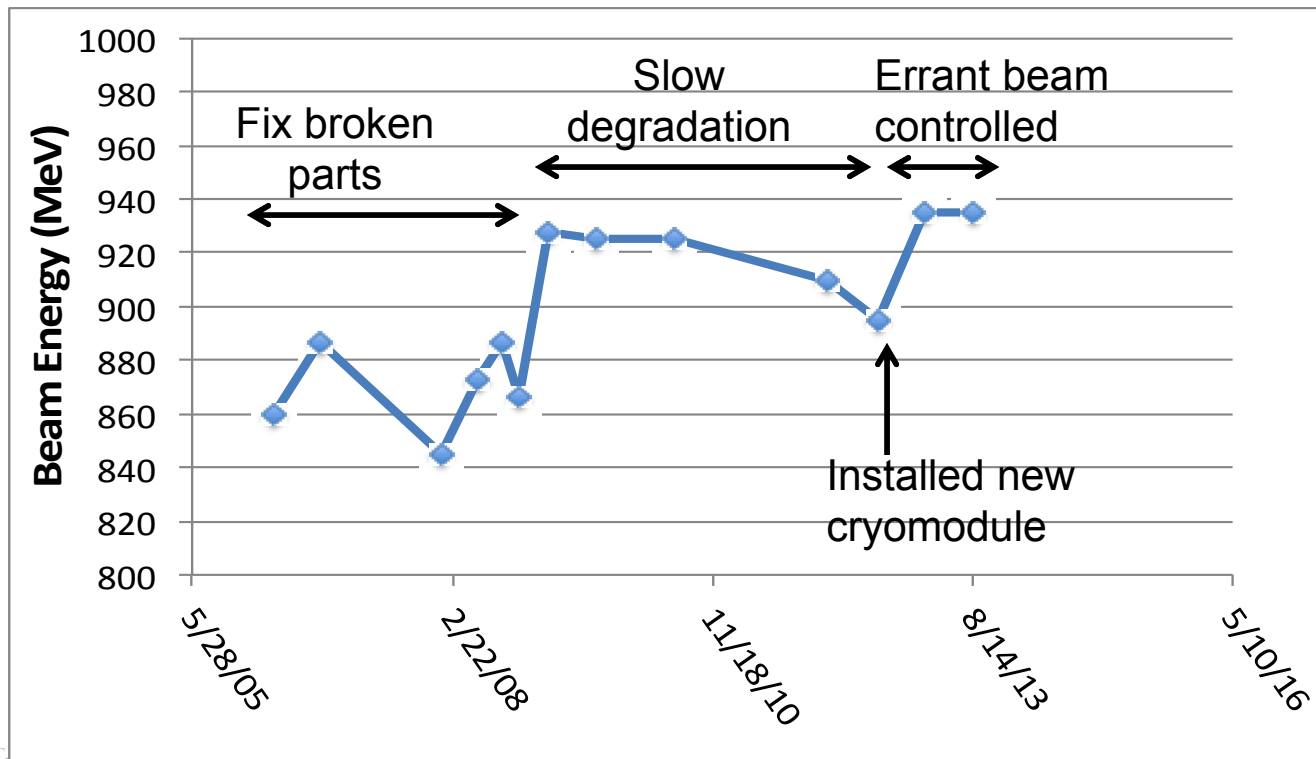
- RFQ is retuned, field is close to nominal again
- Issues with cooling at full duty factor though
 - Unanticipated internal discharge heating
- Beam transmission is still lower than expected (~80%)

Unexpected SCL Gradient Reduction - Errant Beam

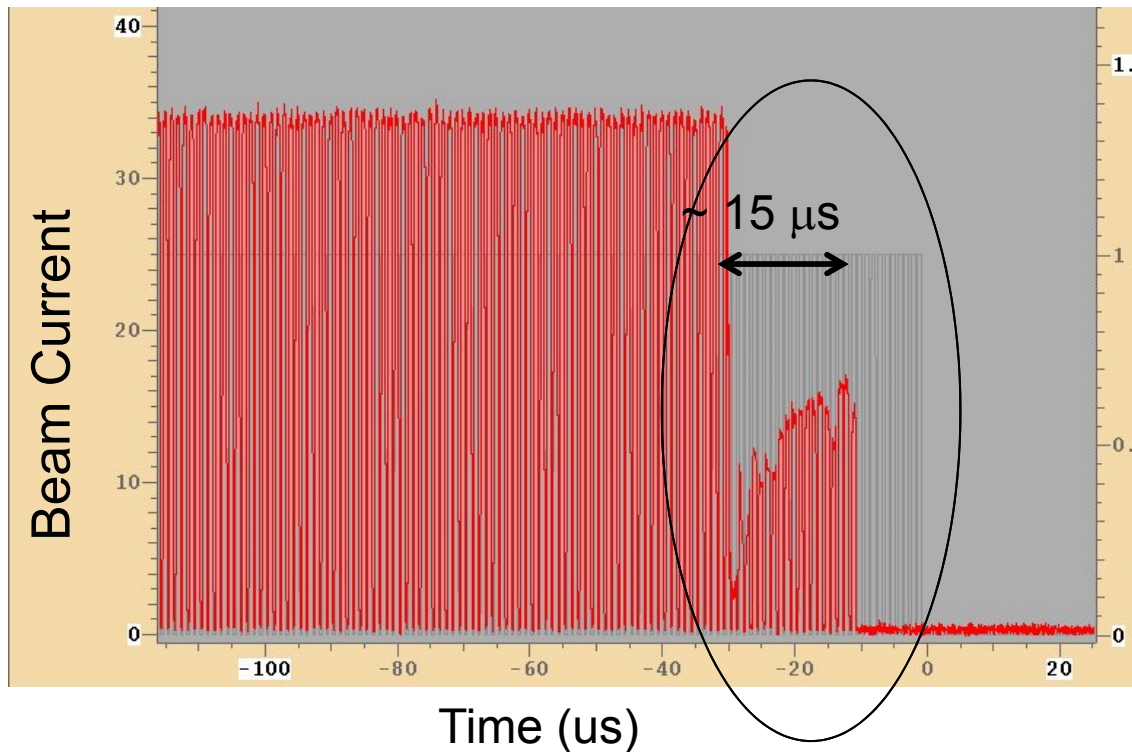
- **Errant beam = beam lost in the Superconducting linac due to upstream equipment malfunction**
- **What causes errant beam?**
 - Ion source misbehavior
 - Upstream warm linac RF truncation
- **Relatively small amounts of repeated beam loss - correlated to reduced SCL performance**

Superconducting RF Performance and Errant Beam

- Machine protection designed to protect against catastrophic damage: melted metal
- Can be more subtle issues
 - Periods of modest beam loss have been followed by cavity performance degradation
 - Transient beam pipe heating induced gas migration ?

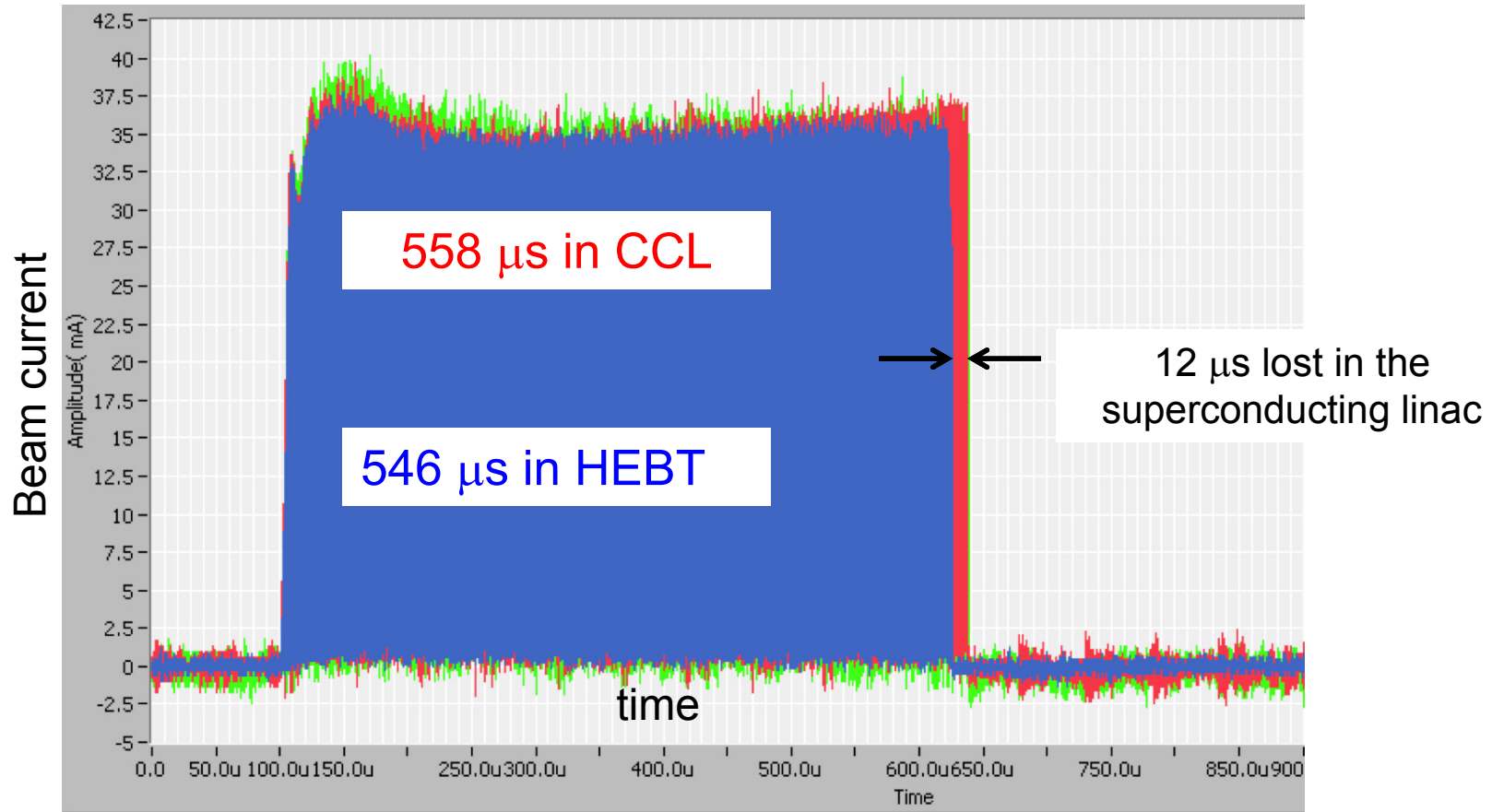


Errant Beam from Ion Source



- Degraded beam current during a pulse
 - Reduced beam loading causes feed forward to overdrive RF, resulting in large beam loss

Errant Beam from RF



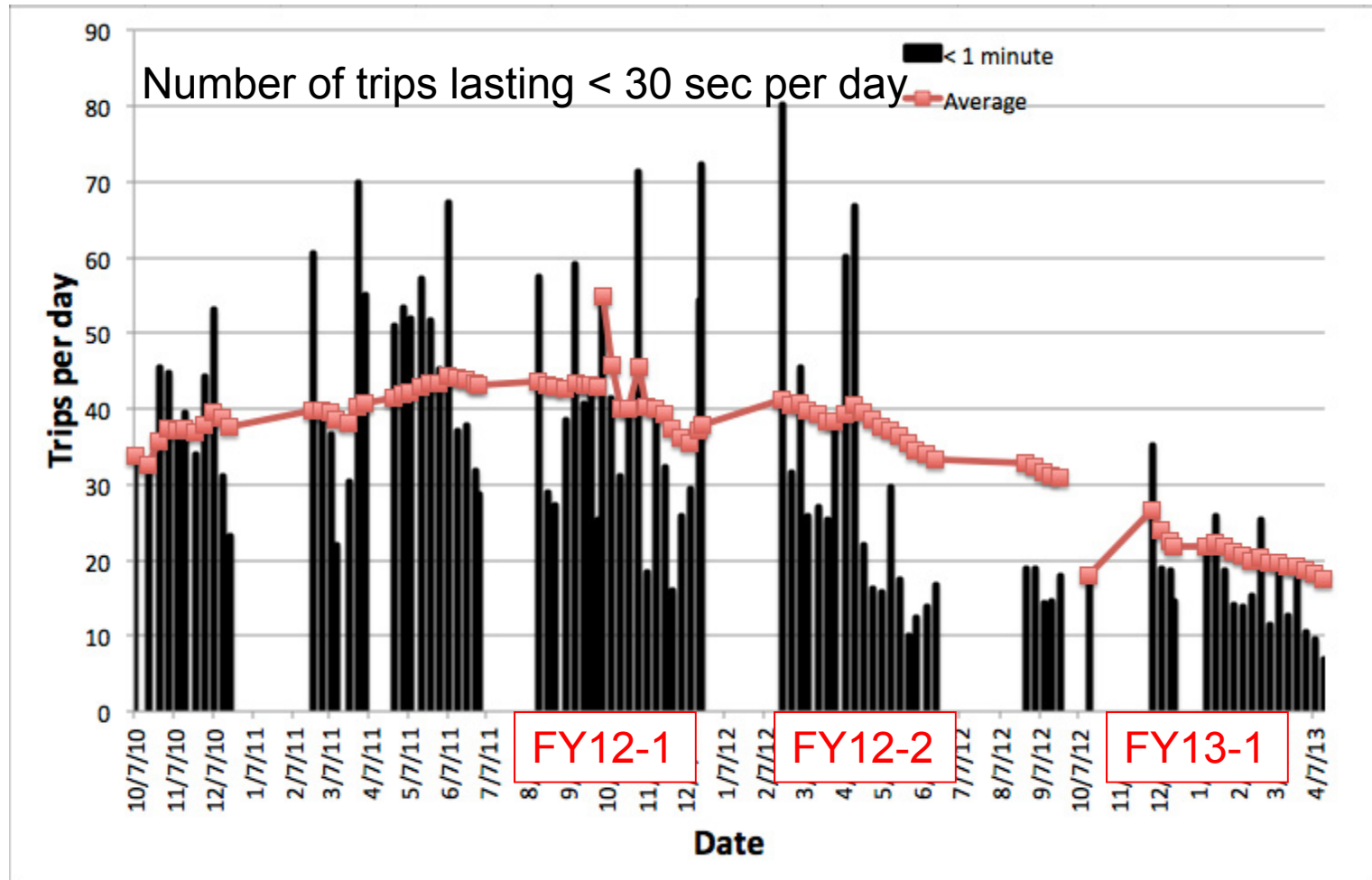
- Truncated warm linac RF pulse due to arc / discharge
 - Beam accelerated before pulse termination is lost in the superconducting linac section

Response to Mitigate Errant Beam Effects

- Identify the causes of beam loss in the SCL
 - 85% RF, 10% Ion source, 5% other
- Reduce the beam shut-off time when one of these events happens
 - “Debug” MPS hardware delays
 - Require operator restart after a single RF arc
- Reduce RF truncation event occurrences
 - Improved vacuum procedures
 - Stringent RF conditioning

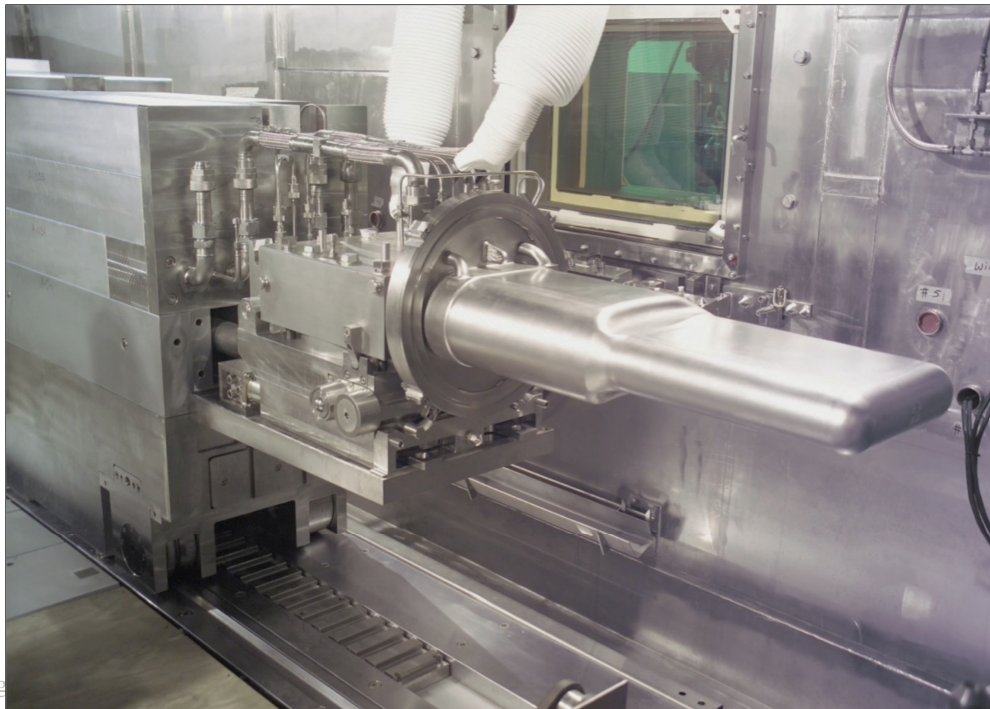
Errant beam trip rates are reduced / SCL Gradient / beam energy stabilized

- By May 2012
 - < 15 errant beam pulses per day



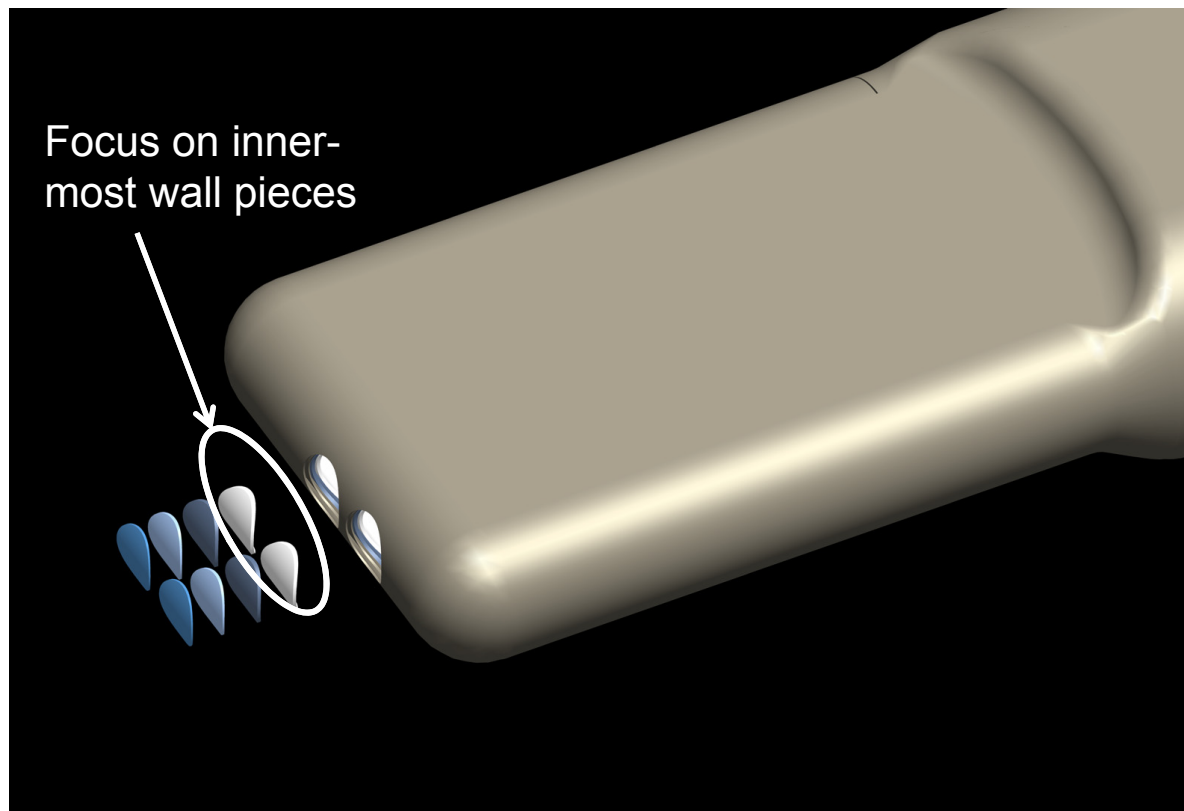
The Business End Experience: Target Woes

- Pulsed load on the target is especially challenging
- Use a liquid target design (Hg)
 - 17 kJ in 700 ns @ 60 Hz
 - Induces cavitation damage

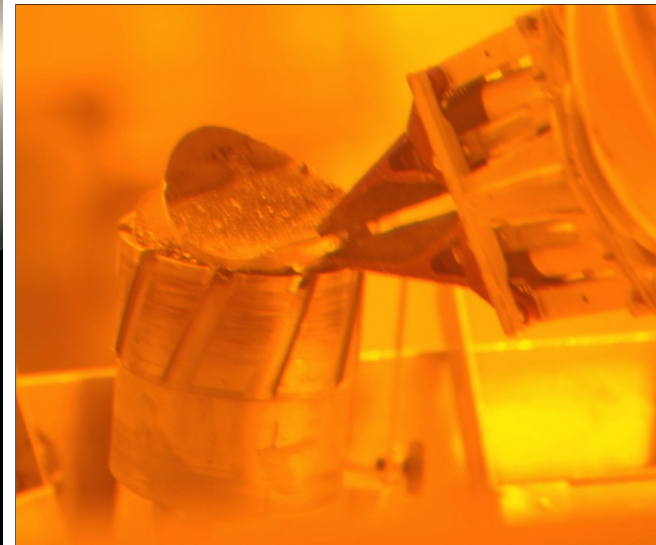


Target Post Irradiation Examination: *key to understanding lifetime*

- Target vessel is multi-walled (4 layers)
- Cut-out circular pieces from used targets to examine damage



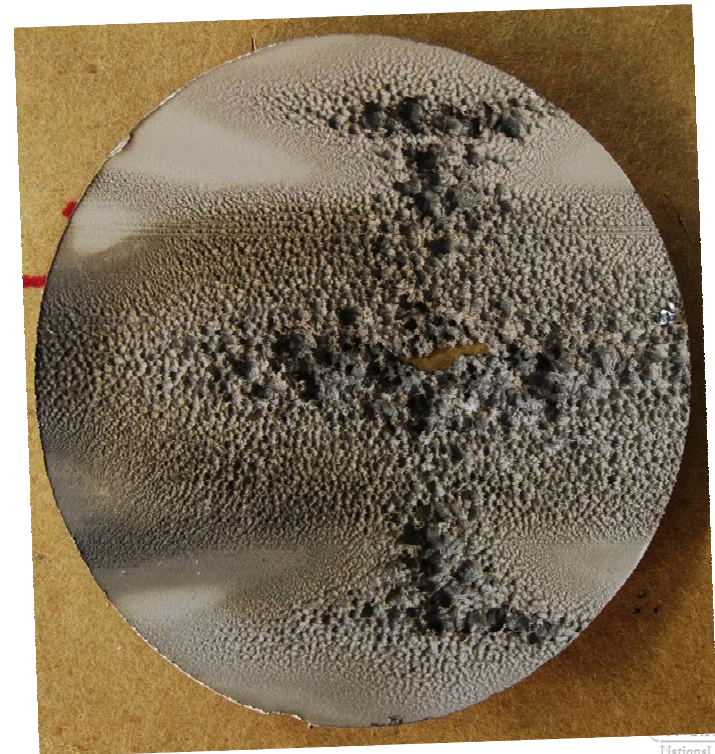
“Gizmo” to cut out pieces



Target Post Irradiation Examination: *key to understanding lifetime*

- Clear indication of cavitation damage
 - Complete “cut-through along centerline
 - Damage is much “sharper” than beam deposition
 - Repeatable

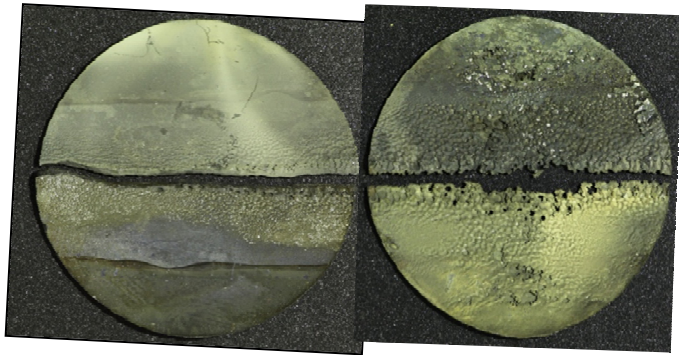
**Target 1 – Inner window
center cut**



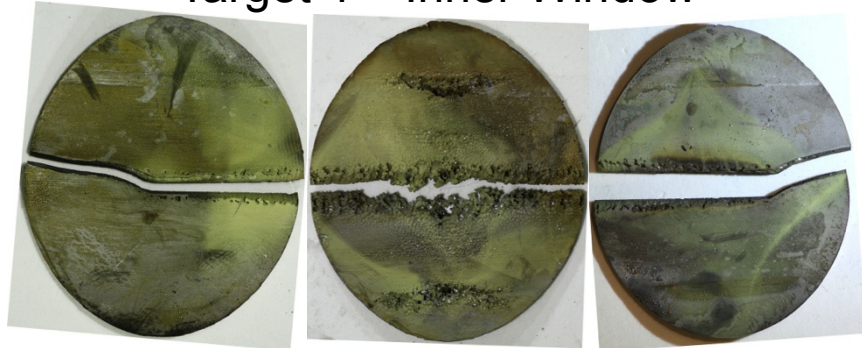
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Target 2 – Inner Window



Target 4 – Inner Window

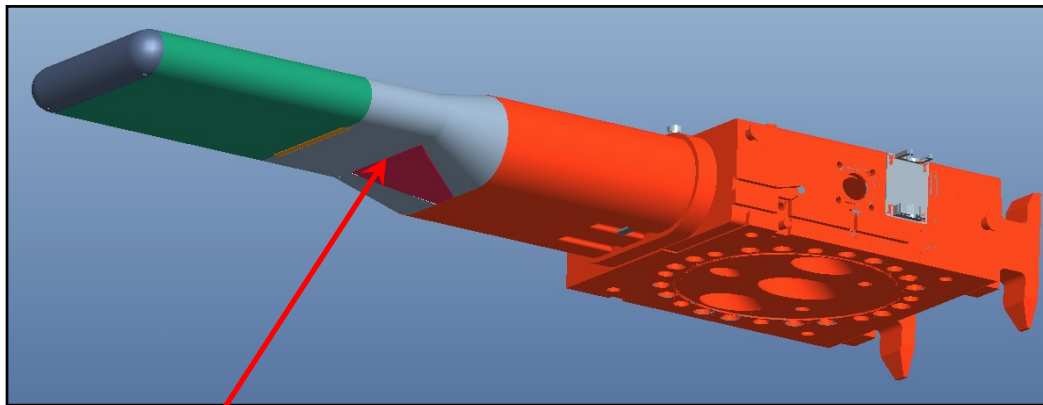


Target 1 – Inner window
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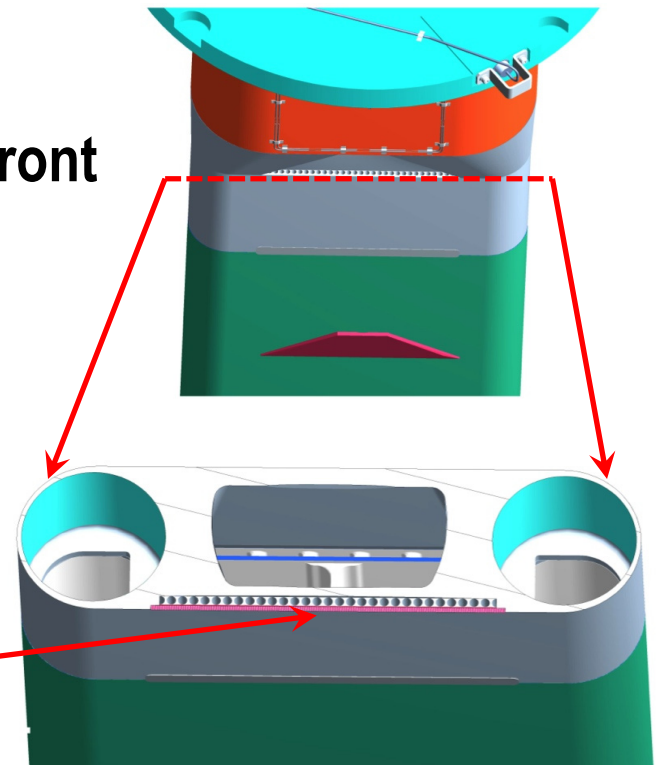


Target Failures in Fall of 2012

- Two successive failures resulted in a target supply shortage
 - Target 6 after ~ 600 hours
 - Target 7 after ~ 100 hours
- Not cavitation induced
 - Poor weld, relatively far from the target front



Targets 6 and 7
Leak Location



Target Maintenance / Replacement is with Remote Handling

Target control

room



Manipulator Gallery



- Not hands on maintenance
- “Improvising capability” is limited

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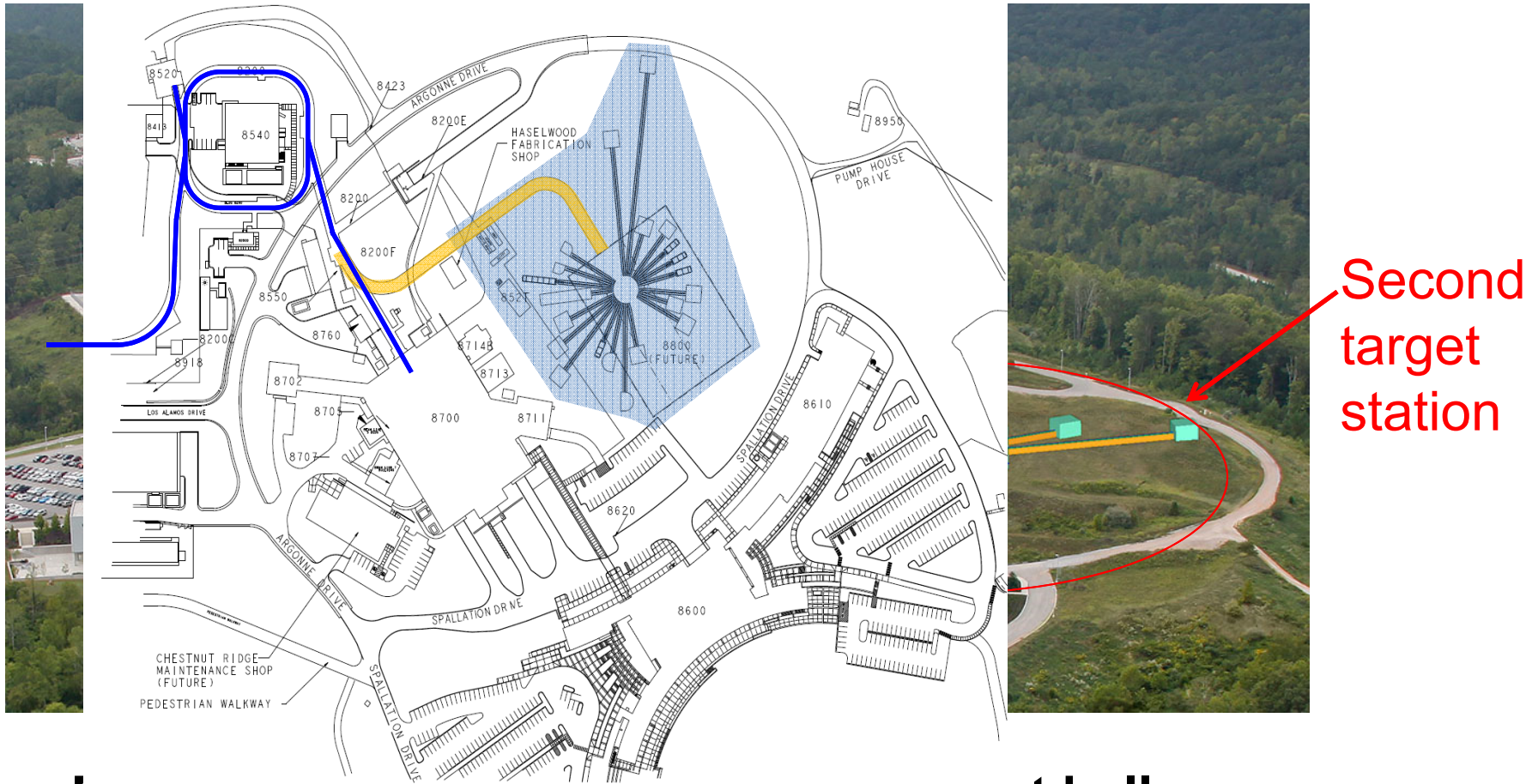
Looking Forward: Second Target Station



Second
target
station

- Provisions for an additional instrument hall were considered in SNS construction
 - Short pulse, long wavelength station under consideration

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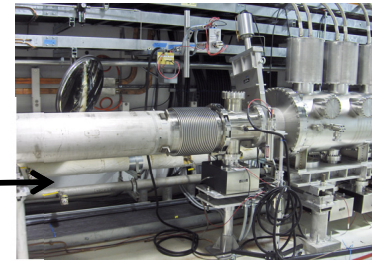
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SNS STS Accelerator Upgrades

- **Increase beam power to ~ 2.5 MW**
 - **Beam energy increase from 930 MeV to 1.3 GeV**
 - Additional room in linac tunnel
 - **Beam current increase from 36 mA to 48 mA (peak)**
 - RF implications – beam loading
 - **Use constant RF Power SCL gradient profile**
 - Some cavities will be run below max. gradient
- **2 MW to the first target (50 Hz), 0.5 MW to the 2nd target (10 Hz)**

SNS STS Accelerator Impact Areas

Tunnel: fill in empty drift sections with cryomodules



Extraction: fill in empty space with kickers

Front-End LINAC



Klystron gallery:
fill in area provided with high power RF equipment

High Power Operational Proton Accelerators

Facility	Power (kW)	Energy (GeV)	Time Structure	Accelerator Type
TRIUMF	100	0.52	CW, 23 MHz	cyclotron
LANSCe area A	80-120	0.8	120 Hz	linac
ISIS Present	200	0.8	40 Hz to TS-1 10 Hz to TS-2	70 MeV H ⁻ linac + RCS
J-PARC MR (FX)	240	30	0.4 Hz x 5 μ s	3 GeV into RCS
J-PARC RCS	300	3	25 Hz x 1 μ s	181 MeV linac + RCS
FNAL MI	400	120	9.4 μ s every 2.2 s	Linac + RCS
CERN SPS	470	400	4.4 s cycle length	linac + 2 stage RCS
SNS	1,000	0.94	60 Hz	linac + accumulator
PSI	1,300	0.59	CW, 50MHz	2 stage cyclotron

- Two operational MW facilities
- 7 facilities with order of 100's kW
- Wide variety of accelerator types

Planned Power Increases - Existing Machines

> 500 kW, development effort ongoing

Facility	Power (kW)	Energy (GeV)	Time Structure	Accelerator Type
FNAL MI + ANU	700	120	9.4 us every 1.33 s	Linac + RCS
J-PARC MR (FX)	750	30	0.4 Hz x 5 us	3 GeV into RCS
CERN SPS	750	400	4.4 s cycle length	linac + 2 stage RCS
LANSCe area A	800	0.8	120 Hz x 625 us	linac
J-PARC RCS	1000	3	25 Hz x 1 us	181 MeV linac + RCS
SNS	2500	1.3	60 Hz	linac + accumulator

- In the 5-10 year time scale: several MW class accelerators expected

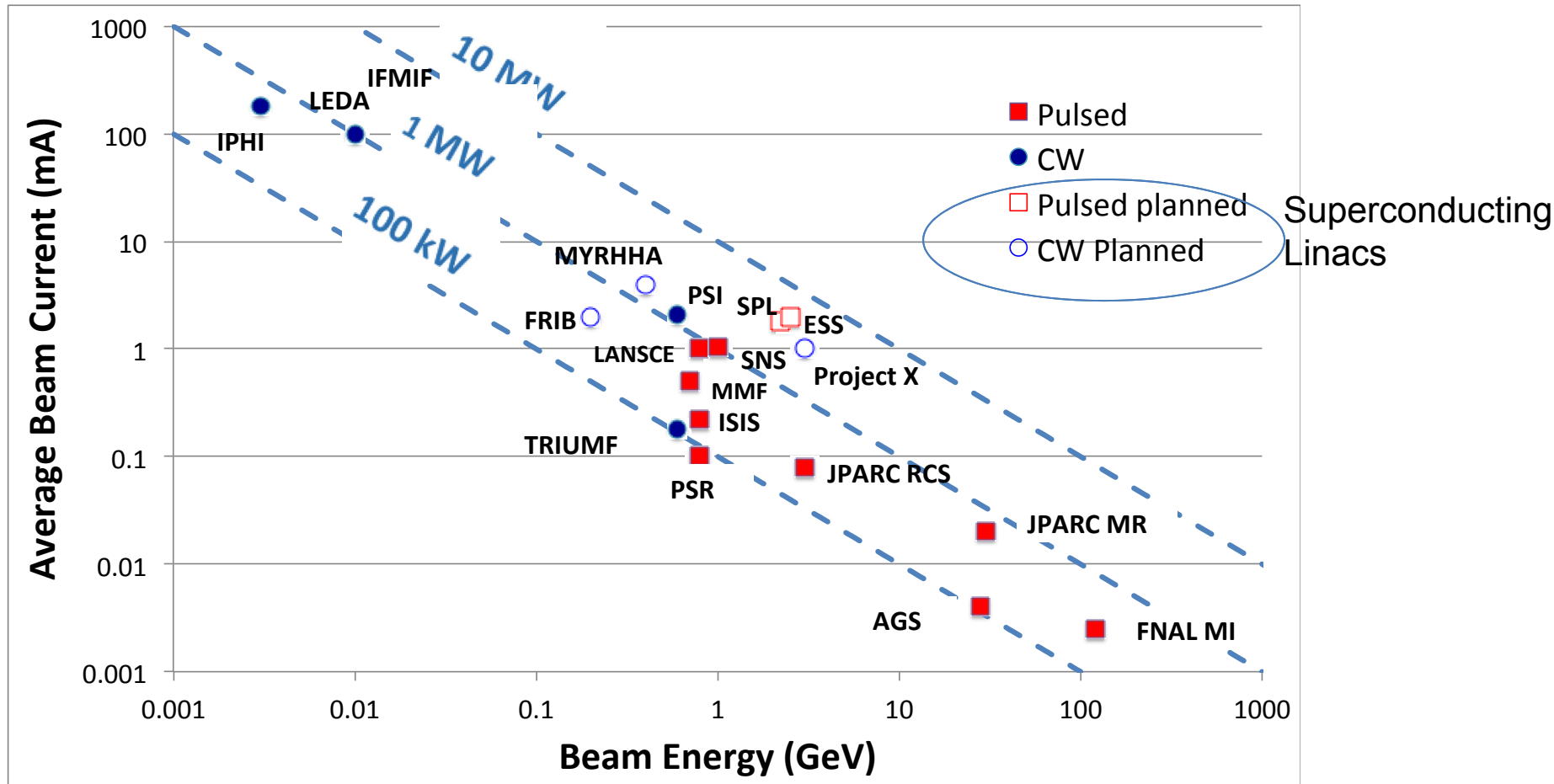
Future High Power Proton Machines

> 1 MW, development or design effort ongoing

Facility	Power (MW)	Energy (GeV)	Time Structure	Accelerator Type
ESS	5	2.0	50 Hz x 2.5 ms	SRF linac + accumulator
CERN with SPL	4	5	50 Hz x 6 bunches	SRF linac + RCS
Project-X (stage 1+2)	3	3	CW linac /accumulator	CW SRF linac + accumulator
Project-X (stage 3)	2.3	60-120	10^{-5} duty factor	+ pulsed 8 GeV SRF linac + RCS
MYRRHHA	2.4	0.6	CW	SRF linac
Daeδalus	3	0.8	60 Hz	2 stage cyclotron

- Superconducting RF linacs play a prominent role

High Power Accelerator History



- Relevant accelerators with ~ MW beam experience
 - PSI: 600 MeV cyclotron, 1.3 MW
 - SNS 925 MeV superconducting linac , 1 MW
 - LANSCCE: 800 MeV copper linac, 800 kW

R&D Areas for Multi-MW Applications

- **Target survivability / robustness**
 - Uncertainty in material properties
- **Beam loss mitigation**
 - Simulation not good enough to predict losses at this level
 - Smart scraping
- **Reliability**
 - For user facility support and ADS applications
- **Beam chopping**
 - User defined structures, multiple targets,
- **Charge exchange injection**
 - Laser stripping

Summary

- **SNS is running above 1 MW, with 1.4 MW demonstration**
 - Some issues, but not so far we are not limited by the accelerator
 - Remarkably flexible machine
- **There are several MW class upgrades and new facilities on the horizon**
 - SRF linacs are a common choice for high power applications